

Problem 292: Compressed Transmission Decoder

Difficulty: Medium

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Problem Background

Lockheed Martin Space designs a wide range of spacecraft for varying purposes, and they all transmit large amounts of data back to Earth. In order to reduce the bandwidth of these messages, data streams are often compressed using a simple encoding prior to being transmitted.

Engineers working on the Orion spacecraft (to be used with NASA's Artemis missions to the Moon) are working on a new compression scheme, and they need your help to test it out.

Problem Description

The compression algorithm identifies repeating sections of data and trims them down in the compressed string. You must decode strings created using this algorithm and reprint the original message.

Each compressed string follows these rules:

- A number followed by a set of parentheses indicates that the content inside the parenthesis is to be repeated that number of times. For example, **3(A)** should be read as **AAA**.
- Parenthesis may be nested within each other; **2(2(A))** is therefore the same as **4(A)**.
- Any content outside of a set of parentheses appears just once; **ABC** is simply **ABC**.
- All numbers in the data stream will be positive integers and will only appear in conjunction with a set of parentheses.

As with all of our problems, we guarantee that all input will be valid and well-formed according to the rules we've outlined in the problem.

Sample Input

The first line of your program's input, **received from the standard input channel**, will contain a positive integer representing the number of test cases. Each test case will include a single line of text consisting of a compressed string as described above. Strings will contain only uppercase letters, parentheses, and numeric digits.

4
3(A)
2(2(A))
A3(BC)
2(A2(B))

Sample Output

For each test case, your program must print a single line containing the uncompressed string.

AAA
AAAA
ABCBCBC
ABBABB